

Nikolai Suhhanov

Medical Simulation Developer | Unity & C# | Former General Surgeon (8+ Years) | Medical XR

Tallinn, Estonia | +372 5599 1961 | suhhanovnikolai@gmail.com | nikolaisuhhanov.com | [LinkedIn](#) | [GitHub](#)

Open to remote work and relocation.

PROFESSIONAL PROFILE

Medical Simulation Developer with more than eight years of clinical experience as a General Surgeon. I combine practical knowledge of surgical anatomy, operating room workflows, instruments and emergency decision-making with Unity and C# development. I build medical education applications and surgical simulation prototypes, with a current focus on Medical XR, procedural interaction and surgical training.

CORE STRENGTHS

Unity & C#

Interactive 2D/3D development

Medical Simulation

Clinical knowledge applied to training design

XR Development

XR Interaction Toolkit and interaction prototyping

Surgical Expertise

Emergency and elective abdominal surgery

DEVELOPMENT EXPERIENCE

Medical Simulation Developer

Jan 2026 - Present

Personal Projects | Tallinn, Estonia

- Independent development of medical education applications and surgical simulation prototypes using Unity, C# and 3D technologies.
- **XR Adhesiolysis Trainer**
Desktop MVP complete; XR adaptation in progress with Unity XR Interaction Toolkit. Includes grasper and scissors interaction, intestinal manipulation, adhesion tension mechanics, visual feedback and cutting logic.
- **Appendectomy Training - Practical Simulator**
Interactive 3D training for the initial stages of open appendectomy with instrument selection and validation, step-by-step workflow and educational feedback. Windows and Android builds.
- **Appendectomy Training - Theoretical Module**
Interactive 3D anatomy, surgical instruments, step-by-step learning and knowledge assessment for the initial surgical approach. Windows and Android builds.
- **Olvar: A Short Journey**
Completed 2D platformer demonstrating general Unity development skills: enemy AI, boss mechanics, dialogue, UI and Windows/WebGL deployment.

Technologies: Unity 6, C#, XR Interaction Toolkit, Git, GitHub, Visual Studio, Android, Windows, WebGL.

CLINICAL EXPERIENCE

General Surgeon

2014 - 2022

Public Healthcare Institutions | Russia

- More than eight years of clinical experience in emergency and elective abdominal and general surgery.
- Independently performed emergency procedures including laparotomies, appendectomies, surgery for intestinal obstruction and other acute abdominal conditions.
- Surgical patient assessment, operative decision-making, postoperative care and multidisciplinary teamwork.
- Practical knowledge of surgical anatomy, instruments and operating room workflows - now applied to medical simulation design.

Selected appointments:

- 2014-2016 Tikhvin District Hospital
- 2016-2018 N. I. Pirogov Clinic of High Medical Technologies, Saint Petersburg State University
- 2018-2020 Emergency Surgeon, City Hospital No. 26, Saint Petersburg
- 2020-2022 E. E. Eichwald Clinic, North-Western State Medical University named after I. I. Mechnikov

EDUCATION & MEDICAL TRAINING

Bachelor of Arts (BA), Web Technologies

Sep 2025 - Jun 2028 (expected)

Estonian Entrepreneurship University of Applied Sciences / EEK Mainor | Tallinn, Estonia

- Current studies include software development, web technologies and user experience design.

Doctor of Medicine (MD), Medicine

Sep 2006 - Aug 2012

Saint Petersburg State Pediatric Medical University

- Medical education forming the clinical foundation for later surgical practice and current medical simulation work.

Postgraduate General Surgery Training

2012 - 2015

- Internship in General Surgery (2012-2013) and Residency in General Surgery (2013-2015).

TECHNICAL SKILLS

Development: Unity 6, C#, Object-Oriented Programming (OOP), debugging, 2D/3D interaction logic

XR / Simulation: XR Interaction Toolkit, medical simulation, surgical education, procedural prototyping

Tools: Git, GitHub, Visual Studio

Platforms: Windows, Android, WebGL

Web: HTML5, CSS3, Bootstrap, basic UX/UI

MEDICAL EXPERTISE

- General Surgery and Emergency Surgery
- Abdominal Surgery and acute surgical conditions
- Surgical Oncology qualification
- Human Anatomy, patient assessment and postoperative care
- Operating room workflows, surgical instruments and surgical education

ADDITIONAL EMPLOYMENT

Cook

Jul 2023 - Present

Park Inn by Radisson Meriton Conference & Spa Hotel Tallinn | Tallinn, Estonia

- Current employment while pursuing a Web Technologies degree and transitioning into medical software development.

LANGUAGES

Russian Native

English B1-B2

Estonian B1

PORTFOLIO & PROFESSIONAL LINKS

Portfolio: nikolaisuhhanov.com

GitHub: github.com/NikolaiSuhhanov

LinkedIn: linkedin.com/in/nikolaisuhhanov

Project demos, medical credentials and professional references are available on the portfolio website.